

Contact

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(LinkedIn)

Top Skills

Plastic SCM

AI Software Development

Cursor AI

Languages

English (Professional Working)

French (Native or Bilingual)

Sébastien Rombaults

Senior Software Engineer, Tools

Greater Paris Metropolitan Region

Summary

Senior Software Programmer working with the Unity Editor and Unreal Engine. Strong engineering professional skills in C++, Build Systems, CI/CD & VCS (Git, Perforce, Plastic SCM/Unity Version Control), Linux, Python, CMake, AI engineering (SKILLS for Claude Code & Cursor)...I have recently been focusing heavily on AI-assisted software engineering: implementing Claude Code skills in our repository to help coding agents follow the team development workflow, project architecture, coding standards, and code review practices.

Experience

Unity

Senior Software Engineer

February 2022 - Present (4 years 5 months)

Paris

I am part of the VCS Tech team, developing Unity Version Control (Plastic SCM) in multiple Engines (focusing on Unity), tools, and services.

- I was hired as the owner and original developer of the Unity Version Control plugin for Unreal Engine 5 (C++)
- I have been focusing lately on the development of the Unity Version Control package for the Unity Editor (C#)
- I handled many of the Tools & CI/CD Integration work in Github related to Unity infrastructure and package release pipeline.

I have been using Cursor (AI IDE) intensively for nearly a year, and then Claude Code:

- I worked on an experimental Perforce Package using the UI Toolkit, leveraging Cursor AI to speed up the prototype (C#).
- I spearheaded the AI effort in our team, to show, teach, and encourage a coordinated team effort: I authored the vast majority of our SKILL files (AI instructions).

- I presented a couple of internal Unity Talks related to the development of Claude Code SKILL.md files

Side notes:

- I handled maintenance of the Perforce Plugin & Version Control integration in the Unity Editor.
- I initially helped with the development of plasticscm.com C# backend & ASP.net frontend in Azure Cloud ("full stack") when I was part of the Integration / Ecosystem team.

Hobby

Open Source Developer (Github)

May 2009 - Present (17 years 2 months)

- C++ wrapper of the SQLite3 library, using CMake, Github Actions etc.
- C++ Logger, C++ HTML 5 generator
- OpenGL3+ & Vulkan tutorial 3D engines
- CodinGame Multiplayer AI competition (The Great Espace)
- C++98 compatible shared_ptr implementation
- C++ Embedded HTTP server using Boost Asio

Freelance

Unreal Engine 4 Plugin Programmer (OSS)

March 2014 - Present (12 years 4 months)

- Plugin Git pour Unreal Engine 4.1, désormais intégré officiellement depuis UE4.7, récompensé par une invitation à participer à la GDC 2016 avec Epic Games.
- Plugin Plastic SCM pour Unreal Engine 4 pour Codice Software.
- Tutoriels et prototypes avec Unreal Engine (ArchViz, jeux, multi, C++ et Blueprints).

Darewise Entertainment

3 years 11 months

Lead Tech and Tools Programmer

February 2020 - February 2022 (2 years 1 month)

Paris Area, France

Lead Tech and Tools Programmer on Unreal Engine 4 with SpatialOS, working on the ambitious MMO Life Beyond, currently in closed alpha:

- Manage the tool programmer, design and help implement scripts and UI for the content pipeline

- Manage the build system engineer, design improvements to Jenkins CI/CD and migrating the scripts to Python using our internal python utils library
- Manage the Backend team, and develop fixes and improvements (C#, Google Cloud) to the matchmaking stack (including Playfab)
- Mentoring programmers, tech artists, and tech designers.
- Development of Blueprint & Python APIs for tech designers & tech artists respectively
- Integration, update, and maintenance of the SpatialOS GDK plugin for Unreal, in collaboration with Improbable tech teams.
- Development of C++ Game Code (network infrastructure, managers, config etc.)
- Intégration of C++ online services (Playfab player account and matchmaking)
- Crash analysis, profiling & optimization, with network/replication issues.
- Development, update, and maintenance of our internal Unreal Engine 4.26 fork, with minor modifications, backport of bugfixes, and improvements to the Perforce Plugin. Pull requests to contribute upstream on Github. Interface with UDN to help track issues or ask for guidance.
- Integration and update of plugins (Houdini Engine, Substance...)

Senior Software Engineer

April 2018 - February 2020 (1 year 11 months)

Paris

Tech, Tools and Gameplay Programmer on Unreal Engine 4 with SpatialOS, working on the ambitious MMO Life Beyond, currently in closed alpha:

- Mentoring of junior programmers, as well as of technical designers and animators.
- Development (with an Intern) of our build system using Jenkins pipelines and Python script.
- Development, update, and maintenance of our internal Unreal Engine 4.22 fork, with minor modifications, backport of bugfixes, and improvements to the Perforce Plugin. Pull requests to contribute upstream on Github. Interface with UDN to help track issues or ask for guidance.
- Integration, update, and maintenance of the SpatialOS GDK plugin for Unreal, in collaboration with Improbable tech teams.
- Development of C++ Game Code (tech & gameplay) network infrastructure, managers, config etc.
- Crash analysis, profiling & optimization, with network/replication issues.
- Development of custom Editor Toolbars and menus

- Development of Blueprint & Python APIs for tech designers & tech artists respectively
- Conception and development of an integrated web server for tooling on the game client.
- Integration with PlayFab online services (authentication and server browser)
- Integration and update of plugins (Houdini Engine, Substance, Unified Interactive Water Syst.)
- Operation of a live server

ENGIE INEO Systrans

12 years

R&D Embedded Software Programmer

May 2009 - April 2018 (9 years)

Achères, Ile de France

- Design, Tests Driven Development and maintenance of the new main product of the company, within a team of 5 developers inside a bigger project of ~30 people
 - Developed our C++ framework, many feature modules, and our test engine
 - Developed CD/CI tools to support it, running parallel builds and tests on multiple platforms
 - Migrated from Subversion/SVN to Git, then leveraging Gitlab & Gitlab CI
 - Help co-manage the team, mentor junior programmers, inform external teams
 - Web tools, VoIP, Android application
 - Intranet tools and some simple web developments
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- Conception, développement dirigés par les tests (TDD), et déploiement du nouveau système d'Aide à l'Exploitation et d'Information Voyageurs (SAEIV) des transports en commun.
 - Formation et encadrement en équipe de 2 à 5 personnes (Agile SCRUM) .
 - Mise en place du framework C++ et des couches d'abstraction de l'OS, des bibliothèques C/C++, portages (WinCE, Linux, Android, OpenAT), du build-système, scripts et hooks git.
 - Conception et réalisation du moteur de scénarios de tests automatiques, du séquenceur au cœur de nos applications,.
 - Applications Android
 - Service de VoIP Protocol Buffer sur TCP/IP
 - Serveur HTTP de maintenance Web embarquée
 - Mise en places des outils de l'intranet : portail custom, MediaWiki, Mantis... et du workflow de développement Git, Gitlab/Gitlab CI, Jenkins... et documentation de référence associée.

Lead low-level software programmer

May 2006 - May 2009 (3 years 1 month)

Achères, Ile de France

- Low-level programming to support new hardware products, drivers, and micro-controller programming
- Responsable des développements bas niveau sur le projet de renouvellement de la gamme de matériel embarqué au sein d'une équipe projet de 4 personnes.
- Conception et développement du logiciel C++ d'orchestration de nos applications embarqués (services TCP/IP et interface Web).
- Développement de drivers C et de noyaux Windows CE pour cartes ARM et x86.

Renault

Assist. Archi. Diag. & Modes Dégradés Contrôle Moteur

May 2005 - May 2006 (1 year 1 month)

- Failure mode analysis & on board diagnosis of the Engine computer
- Architecture et diagnostic des modes dégradés du calculateur moteur (Renault Lardy) projet EM2010.

Eurilogic

Ingénieur Développement Logiciel Embarqué

August 2003 - May 2006 (2 years 10 months)

Low-level Software Programmer:

- FPGA VHDL design & C program of an embedded system for onboard computers for buses, including a microcontroller, a sound system, I/Os.
- WindowsCE driver programming and integration of Board Support Packages

Plusieurs missions de consultant en informatique :

- Conception et développement d'une carte à base de système électronique programmable (FPGA) en VHDL intégrant un micro-contrôleurs, développement du logiciel en C sans OS.
- Développement de logiciels embarqués au sein d'une équipe projet de 10 personnes, portages de BSP Windows CE, de drivers Windows XPE, développement ARM Xscale.

Dassault Aviation

Software Engineer Internship

February 2003 - July 2003 (6 months)

Saint-Cloud

Engineering Internship - Embedded systems

- Implementing a System On Ship on an FPGA including a RISC processor (VHDL) and a C program
- Proof of Concept of a video signal filtering using simulation of the FPGA

Stage Ingénieur de fin d'études - Architecture des systèmes Temps Réels

- Etude d'un System-On-Chip sur composant électroniques programmables FPGA (VHDL) avec cœur RISC programmé en C.
- Conception, développement, simulation et mise au point d'un filtrage d'image vidéo en temps réel, pour affichage sur visu d'avion de combat (picture in picture).

Education

Isep - École d'ingénieurs du numérique

Ingénieur, Architecture des systèmes embarqués / Embedded systems · (2000 - 2003)

Lycée Pothier, Orléans

Classes préparatoires aux grandes écoles, Maths, Physique et Sciences de l'Ingénieur · (1998 - 2000)

Lycée Pothier, Orléans

Baccalauréat, Scientifique · (1997 - 1998)